
THE TOPOLOGICAL FOOTPRINT OF THE UNKNOWN UNKNOWN

*Field Monism, the Scope Argument, and a Trisductive
Residual-Orthogonality Test for Unmapped Covariates*

Trisductive Analysis

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1. ABSTRACT

Standard probabilistic and Bayesian verification architectures remain structurally vulnerable to the unknown unknown: a latent threat or unmapped covariate that resides outside the operator's predefined dictionary. The vulnerability persists because prevailing models treat unknown variables as informational voids rather than physical actors bound by thermodynamic and topological law. This paper carries one load-bearing thesis and one operational instrument, kept structurally separate so the thesis does not depend on the instrument. The thesis is a scope argument: any threat that actuates against a system must leave a persistent distinguishable change in the target, must carry structure, and must establish a boundary relative to the target, and therefore enters the system's own triaxial domain by the act of acting. The energetic requirement rests on the cost of the transition that produces the change, bounded below in energy and time by the quantum speed limit, not on the energy of the resulting state and not on any erasure cost. It binds whether or not the producing process was reversible. There is no exterior from which a zero-effect intervention launches. Naming was never what placed a threat in scope. Acting is. The instrument is a residual-orthogonality test. After completing the orthogonal subtraction of every known energy-bearing covariate, residual correlation among the three measurement axes signals the presence of an unmapped factor before its identity is known. The instrument detects existence, not identity, and it is a graded statistical test, not a binary lock. It runs in two tiers: a fast linear zero-correlation screen, and a kernel-based joint-independence tier that closes the nonlinear evasion route an adaptive adversary would exploit. We state its construction, including a required common-mode reference channel, its admissibility conditions, and the one regime in which it returns no verdict. The architecture reduces the problem from anticipating every threat to completing one's subtractions and reading the residual. That is a real reduction, not the elimination of omniscience in the strong sense, and we mark the limit plainly rather than conceal it.

2. BACKGROUND AND RATIONALE

The structural vulnerability of standard verification methodology to the unknown unknown is not a failure of processing power. It is a defect in how unknowns are modeled. Risk theory and epistemological validation have historically modeled unknowns as negative space: entities outside the boundary of the known. In probabilistic substrates this assumption creates a blind spot. A verification matrix can validate a false model when the disrupting covariate has simply not been named, because the interference that covariate generates is absorbed into the confidence intervals rather than flagged. Bayesian inference conditions on an evidence architecture it does not itself certify. The prior is underdetermined, and the mechanism runs cleanly on broken inputs.

The 2014 BICEP2 announcement of primordial B-mode polarization is the standard cautionary case, and it must be placed in its correct category, because miscategorizing it is itself a common error. Galactic dust polarization at 353 GHz was a known, modeled foreground class. It was not outside anyone's conceptual dictionary. The BICEP2 result failed because that known foreground was under-subtracted. The detector array and the Keck Array follow-up observed the same sky region and shared the dust covariate completely, theoretical consistency with inflation was an interpretive frame rather than an independent channel, and the multi-frequency data needed to subtract the foreground adequately was not yet in hand. The 2015 joint BICEP2, Keck, and Planck analysis revised the tensor-to-scalar limit downward to consistency with no detection. BICEP2 is therefore a known-unknown under-subtraction, not the titular unknown unknown. It is included here for one reason. It demonstrates the precondition on which the operational test of Section 5 depends. The test certifies nothing unless every known energy-bearing covariate has first been subtracted to closure. BICEP2 is what an incomplete subtraction looks like. The defense against the genuinely unmapped covariate is carried not by this example but by the scope argument of Section 4.

The deeper point dissolves the known-versus-unknown taxonomy for purposes of scope, though not for purposes of bookkeeping. Under a continuous-field ontology, any factor that exerts force carries energy-momentum and occupies the field, whether or not an operator has named it. A passed-clean state on a contaminated model is therefore not a successful illusion produced by an exotic ghost. It is a geometrically incomplete audit. The Planck data was not external to the universe. It completed a boundary the operators had truncated. The barrier yields not to an infinitely expanding dictionary of threats but to the geometry of the field and the discipline of completing one's subtractions.

3. BRIEF LITERATURE REVIEW

Classical approaches to deep uncertainty treat the unknown unknown as an informational ghost rather than a physical actor. Bayesian epistemology manages unknowns through broad priors and iterative updating, but as the reference-class problem shows, priors are underdetermined and the evidence architecture is left uncertified. The subjective-objective fracture remains unclosed.

Taleb's analysis of the Black Swan and fat-tailed kurtosis correctly identifies the fragility of predictive models built on Gaussian assumptions. Antifragility, graceful degradation in the face of the unmapped, is a mitigation strategy. It resigns epistemology to perpetual vulnerability rather than addressing the physics of the unmapped variable.

Gödel's incompleteness theorems and Tarski's undefinability theorem establish ceilings on formal-deductive closure. No sufficiently rich axiomatic system proves its own consistency from within. These are limits on deduction, not on physical detection. The present architecture does not contest them on their own ground. It adds two measurement axes that deduction does not natively carry, an empirical-thermodynamic axis and an epistemic-registrational axis, so that a claim undecidable by deduction inside a formal system can still carry orthogonal warrant from the energy it registers and the boundary it establishes. The formal ceiling stays where it is. The detection logic does not require crossing it.

Every prevailing approach separates the mathematics of inference from the physics of actuation. That separation is the gap a field-monist ontology closes.

4. THE SCOPE ARGUMENT

This section states the load-bearing thesis. It uses no determinant and no statistics. It survives independently of the instrument in Section 5.

A threat is anything that actuates against a target. It produces an effect the target would not otherwise have undergone. Consider what actuation requires. First, the threat must leave a persistent distinguishable change in the target. To affect the target in any way the target carries forward, the threat must move it to a state distinguishable from the one it would otherwise occupy. The anchor here is the transition, not the end state. Two distinguishable states need not differ in energy. Orthogonal states in a degenerate eigenspace are perfectly distinguishable while isoenergetic, so no claim about the energy of the resulting state is available or needed. What is bound is the act of producing the change. Driving the target between non-identical states is a physical transition, and a transition is not free. In the quantum regime the speed limits of Mandelstam and Tamm and of Margolus and Levitin require a nonzero energy spread sustained over nonzero time to reach a distinguishable state. In the classical regime the analog is a force acting over time. Either way, producing a distinguishable change requires a physical interaction carrying nonzero energy-time signature, and a process that produces no change requires none. This holds whether or not the process is logically reversible, because the speed limit binds reversible unitaries identically. A reversible perturbation that nonetheless leaves a persistent change is still an actuation, because reaching the changed state was a bounded transition and the persistent change is the registered distinction it left. Reversibility means the process is undoable, not that its result is absent and not that the transition was free. The only case excluded is a process that leaves no persistent change, returning the target to the state it would have occupied regardless. That is not an undoable threat. It is no threat, because the target is then physically identical to its counterfactual and nothing was done to it. An earlier formulation anchored this requirement first on the Landauer erasure floor and then on an end-state energy difference. Both were wrong. The erasure floor binds only logically irreversible operations, and the end-state difference fails because distinguishable states can be isoenergetic. The correct anchor is the transition bound, which is theorem-grade in the quantum regime, binds the reversible case, and makes no false claim about the energy of the result. The Landauer floor returns later as a sufficient signature for the irreversible subclass, not as the foundation of this leg. Second, the threat must carry structure. An unstructured perturbation is noise, not a threat. To constitute a directed intervention the threat must have organization that selects its effect. Third, the threat must establish a boundary. To act on the target rather than to be the target, the threat must be distinguishable from it, which is a relational, self-versus-target distinction.

These three requirements are not three properties one hopes a threat has. They are the conditions for the word actuate to apply. A candidate that lacks the first has registered nothing and has not acted. A candidate that lacks the second is noise. A candidate that lacks the third is not distinct from the target and so is not acting upon it. Whatever actuates therefore populates an energetic axis, a structural axis, and a relational axis. This is the triaxial structure, reached not by stipulation but by unfolding what actuation means. The reversible-but-persistent perturbation is the instructive case. It populates the energetic axis through the nonzero energy-time signature of the transition that produced the change, the structural axis through the specific transformation it selects, and the relational axis through the coupling by which it reached the target from outside it. It is caught on all three, which is why excluding only the genuinely trace-free process costs the argument nothing.

The consequence for the unknown unknown is immediate. There is no exterior to the field from which a threat launches a zero-cost, structureless, boundaryless intervention, because such a launch would not be an actuation. The moment a threat actually affects a system, it has registered, organized, and bounded itself, and by doing so it has entered the system's own triaxial domain. It is in scope. It did not need to be named in advance, because naming was never the operation that placed it in scope. Acting was. This is why the defense does not require predictive omniscience. The threat supplies its own admission ticket by acting.

Two honest boundaries on this argument must be stated. The scope claim is exact about the field and bounded about the volume. Nothing that leaves a persistent effect actuates from outside the field, because producing that effect is a bounded transition carrying nonzero energy-time signature, which is an in-field physical interaction, and the exactness rests on that transition bound rather than on any erasure cost or any claim about the energy of the resulting state. But an operator audits a defined operational volume, not the entire field, and a threat may actuate through a channel the operator did not include in that volume. The scope argument guarantees the threat leaves a triaxial signature. It does not guarantee the operator's chosen volume contains the channel that carries it. Section 5 addresses what can be read inside a chosen volume. The second boundary. The argument establishes that a signature necessarily exists. It does not by itself establish that the signature is readable against background or computable in tractable time. The existence of the mark and the readability of the mark are different claims. The scope argument owns the first. The instrument attempts the second, within stated limits.

5. THE OPERATIONAL TEST

The instrument is a test for residual orthogonality. It is built on standard multivariate statistics, and it is graded, not binary. We construct it explicitly, because a determinant computed on an undefined object certifies nothing.

Construction of the measurement object. Fix an operational volume and a set of N independent measurement channels over it, indexed j from 1 to N . A channel is a single observation event or sensor stream that returns a reading on each of the three axes. The measurement matrix M is three rows by N columns. Because the paper's own rule is that a determinant computed on an undefined object certifies nothing, each row is given an explicit observable and unit rather than left nominal. Row 1, the structural axis, is a per-channel structural reading: the organizational complexity of channel j 's reading, for instance its description length or the local defect count in a fixed neighborhood of the channel, a dimensionless count that varies from channel to channel. It is deliberately a per-channel quantity and deliberately not the volume-level boundary winding of the crossing construction below. That winding is a single integer for the whole volume, so it would be constant across the N channels, carry zero across-channel variance, and be annihilated by the per-row normalization, exactly the common-mode effect this paper flags elsewhere. The volume winding belongs to the separate boundary-crossing detector, not to this per-channel row, and conflating the two would collapse the structural axis to a constant and reduce the determinant to a two-by-two problem. Row 2, the energetic axis, is a transition-cost reading: the energy-time or action budget of the driving that produced the channel's registered change, the quantity the quantum speed limit bounds below, carried in units of action. This is the same observable the scope argument's energetic leg invokes, and it is deliberately not end-state dissipation, because a logically reversible but persistent actuation has a nonzero transition action while dissipating nothing; instrumenting row 2 as dissipation would make the instrument blind to exactly the reversible class Section 4 shows is in scope. Thermodynamic dissipation in joules, the Landauer quantity, is retained only as an optional supplementary channel sensitive to the irreversible subclass, the channel Prediction 3 probes, never as the energetic axis itself. Row 3, the relational axis, is a per-channel relational reading: a coupling or mutual-information contrast between channel j and the target baseline, carried in bits. Here against the target means relative to the target, the self-versus-target relation of the third condition, and not a distinction imprinted in the target, which would be a change and would belong to the energetic transition cost of row two. Read this way the three rows map one-to-one onto the three conditions of Section 4: row 1 to structure, row 2 to the transition cost of the change, row 3 to the self-versus-target relation. The instrument's claim to read the triaxial signature is then realized by construction rather than asserted. Each row is then normalized by its own mean and standard deviation across the N channels, so M -

$\tilde{M}$ has rows of zero mean and unit variance. This per-row normalization is what makes heterogeneous quantities commensurable. A row of action budgets, a row of structural-complexity counts, and a row of relational bits all become unitless z-scored vectors, and what the test compares is their pattern of covariation across channels, not their raw magnitudes. The shared column index j is the sample space. Without it none of the following objects exist. With it they are standard.

The common-mode reference condition. One consequence of working with centered data must be made a construction requirement rather than left implicit. A correlation matrix is computed on mean-removed rows by definition, so a perturbation that is uniform across all channels, a common-mode offset that adds the same amount to every channel of an axis, carries zero across-channel variance and is invisible to the test. This is not a defect peculiar to z-scoring. It is intrinsic to any covariance-based detector, because covariation across the channel index is the detection substrate and a channel-uniform term contributes none. To close the blind spot, the channel set must include at least one genuinely decoupled known-null reference channel, a vacuum or baseline channel j_0 held at a fixed reading independent of the threat. A common-mode offset that shifts the live channels but spares j_0 is no longer uniform across the full set. It becomes a measurable differential against the fixed reference, re-enters the across-channel variance, and is restored to view. This is the standard logic of a dark frame, a tare, or a lock-in reference. Two honest qualifications hold. The reference must be genuinely uncoupled from the threat, since a perturbation global enough to shift j_0 as well is not cured by it. And a single reference is a minimal lever; multiple references, or references held at distinct baselines, strengthen it.

The correlation object. Define R equals $(1 \text{ over } N) \text{ times } \tilde{M} \text{ times } \tilde{M}^T$. R is the three by three sample correlation matrix of the three axes across the N channels. Its determinant, the generalized variance, lies in the closed interval from 0 to 1. $\det(R)$ equals 1 exactly when the three axes are mutually uncorrelated across channels, the orthogonal case. $\det(R)$ approaches 0 as the axes approach linear dependence, the case where one axis is reconstructable from the other two. Because R is a correlation matrix it is symmetric positive semidefinite, so $\det(R)$ is never negative. The test never asks whether the determinant is negative, which would be a meaningless question. It asks how far below 1 the determinant has fallen and whether that fall is statistically significant.

Subtraction of known covariates. Let $\tilde{C}$ be the matrix whose rows are the known energy-bearing covariates over the same N channels, admitted only if each carries nonzero energy-momentum signature. Psychological or heuristic motives carry no energy-momentum and are not covariates. They are excluded as ill-formed inputs, not subtracted. Form the residual

$M\text{-resid} = M\text{-tilde} \times (I - C\text{-tilde-transpose} \times (C\text{-tilde} \times C\text{-tilde-transpose})^{-1} \times C\text{-tilde})$,

the standard orthogonal-projection residual, which removes from each axis the component lying in the span of the known covariates. Re-normalize the residual rows and form R-resid from them.

The verdict. Test whether R-resid is consistent with mutual orthogonality. The statistic is a test of zero residual correlation, with null R-resid equal to the identity, computed on the generalized variance:

$\chi^2 = -\left[(N - k - 1) - \frac{2p + 5}{6} \right] \times \ln \det(R\text{-resid})$,

with p equal to 3 and $p(p - 1) / 2$ equal to 3 degrees of freedom. This is Bartlett's statistic for the null that the correlation matrix equals the identity, that is, a test of zero residual correlation. The word sphericity is deliberately not attached to it, because in standard usage sphericity names a different null, that a covariance equals a scalar multiple of the identity, which is not the hypothesis R equal to the identity tested here. The degree-of-freedom count follows from the construction without that word: z-scoring pins every diagonal entry of R at 1, so the variances carry no free parameter, only the $p(p - 1) / 2$ off-diagonal correlations remain to be tested, and that count is three for p equal to 3. The leading factor uses $N - k$ rather than N because k covariate dimensions have been projected out and the effective independent sample count on the residual is $N - k$. Using the unadjusted N would overstate the effective sample size and bias the test toward rejection. The correction is conservative. It makes a false anomaly on clean data less likely, which is the direction a detector should err. One caveat on this leading factor must be stated. After projecting out k covariates, the residual correlation entries are not distributed exactly as those of a raw sample correlation matrix from $N - k$ independent draws, because the projection itself induces mild dependence among the residual coordinates. The $N - k - 1$ factor is therefore a first-order correction, not an exact one, and at small $N - k$ the permutation null already used for the second tier is the exact fallback when the chi-squared approximation is in doubt. One further correction is mandatory on real data and is not handled by the $N - k$ adjustment. The chi-squared null assumes the N channels are independent draws. Spatially or temporally correlated sensor streams violate that assumption, reduce the effective number of independent channels below N , and inflate the statistic toward false Anomaly, which bears directly on the 5 sigma claim of Prediction 2. The channel count must therefore be replaced by an effective count estimated from the inter-channel autocorrelation, or, more robustly, the null must be obtained by a block-permutation or block-bootstrap procedure that preserves the dependence structure under resampling. Only after that correction is the reported significance honest on dependent data. A $\det(R\text{-resid})$ near 1 yields a small statistic. The residual axes are orthogonal, no

unmapped factor is detected, and the verdict is [⚡] Sealed. A $\det(\text{R-resid})$ significantly below 1 yields a large statistic. Correlation structure remains among the axes after every known covariate has been subtracted. There are two sources that can produce this, and the test does not distinguish them on its own. The first is a factor not in $C\text{-tilde}$, a genuinely unmapped covariate. The second is a factor that is in $C\text{-tilde}$ but entered with a misspecified functional form. Because the projection removes only the linear span of the covariate rows, a known factor whose true influence is nonlinear is only partially subtracted when $C\text{-tilde}$ encodes a linear proxy of it, and the unremoved nonlinear part shows up as residual structure. An Anomaly verdict therefore reports that residual linear structure remains, attributable either to an omitted factor or to an incompletely modeled included one. Disambiguation is downstream work with a concrete procedure: enrich $C\text{-tilde}$ with nonlinear terms of the suspected known factor and re-run. If the residual structure clears, the cause was misspecification. If it persists, an omitted factor is implicated. The verdict is [X] Anomaly. The anomaly verdict reports the existence of unmapped or under-modeled energy-bearing structure inside the volume. It does not and cannot report the source's identity. That is the precise sense of detection without identification.

Why the test is graded, and why that matters. A binary verdict that flipped only at exact rank-deficiency would be defeated by partial contamination, because a real but partial shared factor lowers $\det(\text{R-resid})$ below 1 without driving it to 0, and a binary flip would read that contaminated case as clean. This is the decisive design choice. The zero-correlation statistic is calibrated to the size of the fall, so a significant-but-partial factor produces a significant statistic and is caught precisely in the regime a binary lock would miss. The earlier instinct to declare a contaminated pass impossible was wrong. A contaminated pass is exactly what a binary test produces, and a graded test is the answer to it, not a denial of it.

Admissibility conditions, stated as conditions to be checked. The projection and the verdict are well defined only when four conditions hold. The channel count exceeds the covariate count, k less than N . The residual space is at least three-dimensional. This requires N minus k minus 1 greater than or equal to 3, equivalently N minus k greater than or equal to 4, not N minus k greater than or equal to 3. The extra unit is the dimension removed by centering: z-scoring places every row in the hyperplane orthogonal to the all-ones vector, of dimension N minus 1, and projecting out the k covariate directions inside that hyperplane leaves the residual rows in a space of dimension N minus k minus 1. Three vectors cannot be linearly independent in fewer than three dimensions, so when N minus k minus 1 falls below 3 the 3 by 3 residual Gram matrix is singular and $\det(\text{R-resid})$ is pinned at 0 by dimension alone. At N minus k equal to 3 the residual space is only two-dimensional and the determinant is forced to 0, manufacturing the exact false Anomaly the guard exists to prevent. The floor is therefore N minus k greater than or equal to 4, which is consistent with the N

minus $k - 1$ already appearing in the leading factor of the statistic. The known-covariate covariance is well conditioned, the condition number of $C\text{-tilde}$ times $C\text{-tilde-transpose}$ below ten to the sixth, so that the inverse actually computed is numerically admissible. This conditioning bound sits on the matrix that is inverted, not on $R\text{-resid}$. A small eigenvalue of $R\text{-resid}$ is the signal the test is built to read. It must never be discarded as noise. When the first three conditions fail, the verdict is [?] Inadmissible, meaning the test could not be evaluated and better-conditioned or higher-dimensional inputs are required. Inadmissible is not a finding about the world. It is a finding about the measurement setup, and it is kept strictly separate from the anomaly signal so that a genuine detection is never relabeled as numerical trouble.

What the linear test checks, and the nonlinear tier that closes its evasion route. The zero-correlation statistic measures linear dependence among the residual axes. Linear independence equals full statistical independence only under joint Gaussianity. In the general case the linear test detects linear residual structure and is blind to a purely nonlinear dependence that leaves all pairwise correlations zero. The evasion is concrete. If a residual axis carries Y equal to X squared with X symmetric about zero, then the covariance of X and Y vanishes while Y is a deterministic function of X , so the linear determinant returns Sealed on a fully dependent pair. An adaptive adversary who knows the test reads only linear structure routes contamination into exactly this regime. The instrument therefore runs in two tiers. The first tier is the linear $\det(R\text{-resid})$ zero-correlation test above, fast and closed-form, sufficient for non-adaptive contamination and as a first-pass screen. The second tier is engaged in adversarial settings and replaces the linear correlation with a dependence measure that is zero only under full independence. Distance correlation between two variables, provided each has finite first moment, lies in the closed interval from 0 to 1 and is zero if and only if the pair is independent, capturing nonlinear functional dependence that correlation misses. Pairwise distance correlations among the three axes may be displayed as a diagnostic, but the verdict does not rest on a matrix of them: pairwise independence is not joint independence, and a matrix of distance correlations is not a generalized-variance object whose determinant carries meaning. The verdict rests on a single joint statistic. Test the joint independence of the three residual axes directly with a kernel joint-independence statistic of the Hilbert-Schmidt type, which maps the axes into a reproducing-kernel space with a characteristic kernel and is zero if and only if the three are mutually independent. The second tier closes the Y equal to X squared evasion and every dependence the chosen kernel can resolve. Two honest constraints distinguish this tier from the first and must not be elided. Its null distribution is not the closed-form chi-squared of the linear test. The generalized-variance and Bartlett construction is specific to the Gaussian linear correlation matrix and does not transfer to a distance-correlation or kernel object, so the second tier obtains its null by permutation, block-structured where the channels are dependent. And it requires a characteristic kernel and a

bandwidth, which are modeling choices, with finite-sample power set by the channel count. The second tier raises the bar to all nonlinear dependence the kernel resolves; it does not grant omniscience against an adversary free to hide outside the kernel's effective sensitivity at finite sample size. Stated this way the nonlinear evasion route is closed by a specified instrument rather than deferred, and the residual limit is named rather than hidden.

The boundary loop and winding. A separate construction addresses threats that cross into the volume rather than sitting statically within it. A threat that does not cross the operational perimeter exerts no force on the interior and is indistinguishable from the null background. A threat that actuates against the interior must cross the boundary. For a continuous field defined on the region, the number of singularities enclosed by a boundary loop is fixed by the winding of the field around that loop, computed from boundary data alone. In the planar case this is the argument principle. The net number of enclosed zeros or poles of an analytic field equals one over two π times the contour integral of the logarithmic derivative around the loop, an integer read entirely from the perimeter without any interior map. This is the rigorous core of detection without interior omniscience, and it is classical. Its faithful generalization to n dimensions is the Brouwer-Kronecker degree, not a soliton charge. For a continuous map f from an n -dimensional region into n -dimensional space, nonvanishing on the boundary, the signed count of interior zeros equals the topological degree of the boundary map f over its modulus, the degree of a map from the bounding sphere to the unit sphere, computed by the Kronecker boundary integral. The planar argument principle is exactly the one-dimensional complex case of this degree. The Brouwer-Kronecker degree is the right apparatus for three reasons. It is genuinely a boundary integral, it genuinely counts enclosed zeros, and it requires only continuity and nonvanishing on the boundary, neither analyticity nor any homotopy-group structure. This last property weakens the analyticity assumption the planar case had carried as a residual condition: the degree-theoretic read-out needs no meromorphy. The Skyrme and Faddeev-Niemi soliton charges are integer topological invariants of field configurations and are cited here only as illustrations that integer charge is a robust feature of field theory; they are bulk integrals of a topological current over the whole domain, not boundary read-outs of an enclosed singularity count, so they are not the generalization of the argument principle and are not load-bearing here. The remaining honest condition is modest and is stated rather than implied away: the construction models the verification field as a continuous nonvanishing map on the boundary and the enclosed threat as an interior zero of that field, and within that model the boundary integral is rigorous.

6. FALSIFIABLE PREDICTIONS

Each prediction states a threshold, a method, an expected outcome, and a null whose confirmation falsifies the claim. Quantitative thresholds are forecasts to be tested, not reported results.

Prediction 1. Substrate drift under partial initialization.

Prediction. A synthetic substrate running the residual test without a per-turn reload of its operating discipline, before any output, shows a false-Sealed rate that increases monotonically with turn count, while a control arm that reloads each turn does not. The divergence is parameter-free: the claim is the monotone separation and its significance, not a fixed percentage. Mechanism: default reflexes retain authority under any overlay, no internal sensor for the drift exists, so it accumulates unless the discipline is reloaded.

Method of confirmation. Cross-substrate longitudinal tracking across multiple vendor models under automated adversarial prompt injection, two arms, reload versus no reload, measuring false-Sealed rate against turn index.

Expected outcome. Monotone rise in the no-reload arm, flat control, significant separation by the tenth turn.

Null hypothesis. No separation between arms across turns, falsifying the per-turn requirement.

Prediction 2. Pre-identification detection in high-energy pipelines.

Prediction. In multi-variable detection pipelines, the residual zero-correlation statistic on raw sensor streams rejects orthogonality at or above 5 sigma when a significant unmapped energy-bearing covariate is present, before that covariate is identified, once the inter-channel dependence correction has been applied so the 5 sigma is honest. The rejection signals that an unaccounted factor exists and fires for partial factors, not only total ones. It does not name the factor; identification is a separate downstream task.

Method of confirmation. Retrospective application to the BICEP2 dataset across the full frequency spectrum, where the dust factor is now known and serves as ground truth, and to next-generation B-mode foreground data from CMB-S4 and the Simons Observatory. Prospective application to direct dark-matter searches, ADMX-G2 and LUX-ZEPLIN.

Expected outcome. The statistic rejects orthogonality weeks to months before standard pipeline diagnostics isolate the identity of the anomalous source.

Null hypothesis. The statistic fails to reject orthogonality despite an active unmapped energy-bearing covariate, falsifying detection without identification.

Prediction 3. Thermodynamic signature of irreversible adversarial registration.

Prediction. Any adversarial intervention that registers an irreversible distinction against the target incurs a thermodynamic cost bounded below by $k_B T \ln 2$ per registration. This prediction probes the supplementary dissipation channel, not the energetic axis itself, which is instrumented as transition action. The bound applies to logically irreversible operations only. A reversible computation that registers nothing produces no irreversible effect and is not an actuation against the target; it sits in the null background rather than escaping the bound. The bound is a floor on irreversible actuation, not a claim that every computation pays it.

Method of confirmation. Since $k_B T \ln 2$ is about 3 times ten to the minus twenty-one joules at room temperature, below current direct thermal resolution against device dissipation, the test is not a spike search. It is a measurement of the monotone correlation between the count of irreversible registrations an attack performs and its net dissipation, holding device platform fixed.

Expected outcome. Net dissipation rises with the count of irreversible registrations, with reversible-only attacks clustering at idle dissipation.

Null hypothesis. An attack performing many irreversible registrations shows no associated dissipation increase, violating the floor and falsifying the actuation-cost claim.

7. DISCUSSION

The architecture moves verification from a psychological state of confidence to a measurement of residual orthogonality after honest subtraction. Two features deserve direct comment, because they were the points on which earlier formulations failed.

The first concerns the contaminated pass. A finite auditor cannot map an entire field and may always miss a latent variable. The correct response is not to deny that a contaminated pass can occur. It can. A binary verdict that flips only at exact rank-deficiency manufactures contaminated passes, because partial contamination lowers the generalized variance without zeroing it. The graded zero-correlation statistic is the answer. It is sensitive to the magnitude of the residual structure, so a significant partial factor produces a significant rejection. What remains genuinely undetectable by this instrument is a residual structure that is purely nonlinear, leaving all linear correlations zero, and a factor that actuates entirely outside the chosen operational volume. These are stated limits, not hidden ones. The nonlinear limit had a sharp adversarial consequence in the linear-only formulation, and it is now closed rather than deferred. Against an adaptive adversary who knows a linear test reads only linear structure, contamination can be routed into the purely nonlinear regime where all pairwise correlations vanish and a linear test returns Sealed on genuinely contaminated data. The second tier of Section 5 closes this route. By replacing the linear correlation with a joint kernel independence test in a reproducing-kernel space, with pairwise distance correlation kept only as a diagnostic, the instrument detects dependence with zero linear correlation, including the Y equal to X squared case the adversary would exploit. What remains is the honest residual of the kernel tier itself, not a conceded surrender: its null is permutation-based rather than closed-form, it requires a characteristic kernel and bandwidth, and its finite-sample power is bounded by the channel count, so an adversary with enough freedom to hide outside the kernel's effective sensitivity at the available sample size is not excluded in the absolute. The door the linear test left open is shut. The remaining gap is the finite resolving power of any real test, stated plainly. The same root cause unifies the two Section 5 caveats: the linearity of the projection under-subtracts a nonlinear known factor, producing the misspecification ambiguity in the Anomaly verdict, and the linearity of the first-tier statistic misses a nonlinear unknown one, producing the evasion route. The second tier addresses the detection side of that root cause directly, and the disambiguation procedure addresses the subtraction side. One root limitation, the linearity of the operation, two consequences, both now carried by specified instruments rather than left as open warnings. The instrument reduces the unknown-unknown problem to completing one's subtractions and reading a dependence statistic on the residual at the tier the threat environment demands. It does not abolish the finitude of the auditor.

The second concerns verdict structure. The test produces three outcomes that do not overlap. Sealed means the residual is consistent with orthogonality. Anomaly means residual correlation remains after complete subtraction, indicating that unmapped or under-modeled energy-bearing structure is present, with the two sources disambiguated downstream per Section 5. Inadmissible means the measurement setup did not permit evaluation. The anomaly outcome is reachable, because it is a significance threshold on a generalized variance that lives strictly between 0 and 1, not a demand that the variance hit exactly 0. The conditioning bound that guards the projection sits on the covariate covariance, the object actually inverted, and never on the residual, so a genuine detection is never swallowed as numerical noise. There is no fourth verdict, and no formal-deductive ceiling is imported as a verdict. Undecidability of the Gödel or Tarski kind is acknowledged at its own layer and handled by the empirical and registrational axes rather than by a hedge between Sealed and Anomaly.

Two distinct self-reference problems must be separated, because an earlier formulation treated them as one solved problem and they are not the same. The first is the target's interior: verifying an enclosed condition without mapping the interior. The boundary construction handles this one. The auditor verifies the perimeter of the volume, the argument principle and its degree-theoretic generalization read enclosed content from the boundary alone, the interior center is never required to verify itself, and the auditor and the target remain operationally distinct across the loop. The second is the auditor's own drift, and the boundary construction does not touch it. Prediction 1 states that a synthetic substrate executing the test degrades silently, its default reflexes retaining authority under any overlay with no internal sensor for the drift, so it cannot certify its own continued fidelity from within and requires an external per-turn reset. That is a genuine self-reference limit on the auditor, not on the target, and it is not bypassed; it is relocated to whoever performs the reset. Calling it bypassed would be the error. The honest statement is that the boundary construction solves target self-reference, while auditor drift is a separate failure handled by external reset, and this costs the thesis nothing, since the thesis already concedes the finitude of the auditor.

The cost is honest and stated. The instrument requires a defined operational volume, a complete subtraction of known energy-bearing covariates, a constructed sample space of measurement channels, a per-turn reset of any synthetic substrate executing it, and the acknowledgment that it reads linear residual structure inside a chosen volume, not nonlinear structure and not what lies outside the volume. The thesis it serves, that nothing actuates from outside the field, is stronger than the instrument and does not depend on it.

8. CONCLUSION

The defense against the unknown unknown is not an infinitely expanding dictionary of named threats. It rests on a single structural fact and one honest instrument. The structural fact is that anything which actuates against a system thereby registers, organizes, and bounds itself, and so enters the system's own triaxial domain by the act of acting. There is no exterior from which a zero-cost intervention launches. The instrument is a residual-orthogonality test. Complete the subtraction of every known energy-bearing covariate, then read the generalized variance of the residual against the orthogonality null. A significant fall reports an unmapped factor's existence, not its identity, and reports it for partial factors as well as total ones. The architecture reduces anticipation of every threat to the discipline of complete subtraction and the reading of a standard statistic, within a stated volume and against linear structure. We name those limits rather than hide them. The threat that acts leaves a signature that the field's geometry guarantees. Reading that signature is bounded work, and we have specified its bounds.

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10. APPENDIX A. FOUNDATIONAL PRINCIPLES

These are stated as physical and mathematical principles, presented as correspondences to the verification problem, with the strength of each correspondence marked honestly. Where a claim is an interpretive mapping rather than a proven identity, it is labeled as such.

Transition-energy floor (the energetic anchor of the scope argument).

The energetic leg of Section 4 is anchored on the cost of the transition that produces a change, not on the energy of the resulting state. It must not be stated as an end-state claim, because distinguishability does not imply an energy difference: orthogonal states in a degenerate eigenspace are perfectly distinguishable while isoenergetic. The defensible claim is that driving a target to a distinguishable state is a physical transition bounded below in energy-time. In the quantum regime the Mandelstam-Tamm and Margolus-Levitin speed limits require a nonzero energy spread sustained over nonzero time to reach an orthogonal state, with the orthogonalization time bounded below by a quantity inversely proportional to the available energy spread. In the classical regime the analog is a force acting over time to move the configuration. A process that produces no change incurs no such bound, correctly, because nothing was transitioned. This anchor is theorem-grade in the quantum regime rather than a bare premise, it binds logically reversible transitions identically since the speed limit is indifferent to logical reversibility, and it makes no false claim about the energy of the end state. It is the corrected replacement for two earlier and mistaken anchors: the Landauer erasure floor, which binds only irreversible operations, and an end-state kinetic-energy difference, which fails on isoenergetic distinguishable states.

Thermodynamic cost of irreversible registration (Landauer, narrow role).

Every logically irreversible operation, paradigmatically the erasure or irreversible recording of a distinction, requires thermodynamic work bounded below by $k_B T \ln 2$. Logically reversible operations carry no such floor. This bound is used in this paper only as a sufficient detection signature for the irreversible subclass of actuations, in Prediction 3. The scope argument of Section 4 does not rest on it. It rests on the transition-energy floor above, which a logically reversible but persistent transition also satisfies. Demoting Landauer from the foundation of the scope argument to a signature for one subclass is what closes the reversible-perturbation objection without narrowing the class of threats the argument covers.

Generalized variance as an orthogonality measure (multivariate statistics). For the three-by-three sample correlation matrix R of the residual axes, $\det(R)$ lies between 0 and 1. It equals 1 exactly under mutual orthogonality and approaches 0 under linear dependence. Because R is symmetric positive

semidefinite, $\det(R)$ is never negative. The only meaningful question is the magnitude of its fall below 1, tested for significance by Bartlett's statistic for the null that the correlation matrix equals the identity, a test of zero residual correlation carrying $p(p-1)/2$ degrees of freedom, three for p equal to 3. The word sphericity is not used for this statistic, since in standard usage that term names the distinct null of a covariance proportional to the identity rather than R equal to the identity. The count follows from the construction: z-scoring fixes the diagonal at 1, so only the off-diagonal correlations are free, and that is three. This is a standard statistic, not a novel construction.

Nonlinear dependence measures for the adversarial tier (distance correlation and kernel independence). Distance correlation between two variables, provided each has finite first moment, lies between 0 and 1 and is zero if and only if the variables are independent, so it registers nonlinear functional dependence that linear correlation cannot see, including the case of zero covariance with full deterministic dependence. The finite-first-moment condition is part of the characterization and is stated here rather than assumed silently. A matrix of pairwise distance correlations is a diagnostic only, since pairwise independence is not joint independence and such a matrix is not a generalized-variance object; the verdict rests on a joint statistic. The Hilbert-Schmidt independence criterion is zero if and only if independent for a characteristic kernel, and its joint extension tests the mutual independence of several variables at once. The second tier of the instrument uses the joint kernel statistic in place of the linear correlation, with significance obtained by permutation, block-structured under inter-channel dependence, rather than by the closed-form chi-squared that is specific to the Gaussian correlation determinant. The dependence is real and standard; the kernel and bandwidth are modeling choices, and finite-sample power is set by the channel count.

Triaxial decomposition (Hodge, as interpretive correspondence). On a compact Riemannian manifold any differential form decomposes uniquely into exact, co-exact, and harmonic components, with no fourth orthogonal subspace. This paper does not claim an identity between the Hodge components and the formal, empirical, and registrational axes. The triaxiality of the verification axes is argued independently in Section 4 from the conditions for actuation: energetic, structural, and relational. The Hodge theorem is cited as a structural precedent for irreducible three-fold orthogonal decomposition in a continuous setting, a correspondence, not a derivation of the axes from the theorem.

Detection of enclosed singularities from boundary data (argument principle and its degree generalization). For an analytic field on a planar region, the net count of enclosed zeros or poles equals one over two pi times the contour integral of the field's logarithmic derivative around the boundary loop, an integer computed entirely from the perimeter. This is the rigorous basis for certifying an interior singularity without an interior map. Its correct

generalization to n dimensions is the Brouwer-Kronecker degree: for a continuous map nonvanishing on the boundary of an n -dimensional region, the signed count of interior zeros equals the degree of the boundary map onto the unit sphere, given by the Kronecker boundary integral, with the planar argument principle as the one-dimensional complex case. This generalization needs only continuity and nonvanishing on the boundary, so it does not require analyticity, which weakens the residual assumption the planar statement had carried. The Skyrme and Faddeev-Niemi soliton charges are integer topological invariants in field theory and are mentioned only as illustrations that integer charge is robust; they are bulk integrals of a topological current, not boundary read-outs of an enclosed count, and are not the generalization used here. The residual condition is the modeling of the verification field as a continuous nonvanishing boundary map with the enclosed threat as an interior zero, stated as a condition rather than an automatic identity.

Continuous-field ontology (substrate monism). The fundamental substrate is taken to be a single continuous field. Discrete entities, including localized actors and bounded threats, are observer-imposed discretizations of continuous gradients. No actor originates from a space outside the field, so no actuating threat is exempt from the field's geometry. This is the ontological premise from which the scope argument of Section 4 proceeds, stated as a premise, not proven here.